Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/09/2006
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Project Title:	Development of a Finger Calorimeter to Indicate General Thermal Status of Astronauts Onboard and During EVA		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced extravehicular activity		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	55455	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	05/27/2003	**End Date:**	04/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Leon, Gloria Rakita (University of Minnesota)		
Grant/Contract No.:	NAG 9 1521		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A specific aim of this project is to design and test a miniaturized heat flux/temperature finger sensor and monitoring system to function as a finger calorimeter, applying experimental data collected in rest and exercise laboratory conditions. Years 1-2 have been devoted to developing and testing different heat flux ring prototypes. In collaboration with the Mini Mitter, Inc. company, a telemetric monitoring system for measuring finger heat flux/temperature is in the process of development and testing, evaluating several variations of heat flux rings. A telemetric finger temperature sensor has been tested with human subjects in different thermal environmental conditions, under varying heat deficit/surpluses in the body. Mini Mitter worked on the completion of the heat flux component of the finger calorimeter that was tested and evaluated in laboratory conditions. Year 3 has focused on adjustment of the physiological range of the sensor for compatibility with space equipment, further development, refinement and testing of the sensor, transmitter and monitoring system, adaptation of the finger calorimeter to the space suit, and ground-based testing. In addition, initial efforts are underway to assess the efficacy of physiologically/anatomically designed warming glove prototypes.		

	Develop a miniaturized heat flux/temperature finger sensor and monitoring system for eventual use in evaluating thermal status of astronauts during EVA. Our prior research demonstrated that temperature/heat flux density from the finger is highly correlated with heat content in the body in the “vasomotor zone”. This finding is the basic rationale for the development of a finger calorimeter. The heat flux/temperature finger sensor and monitoring system will be a key component in the advanced EVA development of an automatic thermal feedback system to control thermal status of astronauts during EVA by maintaining thermal balance through the life support system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Applications of the technology for better monitoring of thermal status of personnel in other types of extreme conditions, such as military, HAZMAT situations. Advances in designing support technologies for human protection.
Task Progress:	In FY 2005, activities focused on 1. collaboration with Mini-Mitter and the University of Minnesota’s Departments of Mechanical Engineering and Electrical Engineering on the development and refinement of the finger calorimeter, including the programming and testing of the Mini-Mitter VitalSense monitor, transmitter and skin temperature sensors in the laboratory, and miniaturization of the transmitter; 2. assessment of the function of the Phase VI glove in cold conditions with and without insulation; 3. evaluation of the operation of the finger calorimeter, Phase VI, and physiologically designed gloves while subjects grasp a highly cold rail; 4. continued studies on the efficacy of different physiologically-based glove designs to provide hand comfort under significant chilling of the rest of the body; 5. studies on the effectiveness of a head/hand bypass in the tubing of our recently developed shortened liquid cooling/warming garment (SLCWG) in prolonging hand comfort during body chilling; 6. work with the University of Minnesota’s Department of Design, Housing, and Apparel on improving the conformal fit of the garment and gloves. 1. Finger calorimeter development A dermal temperature sensor and dermal heat flux sensor for use on Mini Mitter’s VitalSense® Integrated Physiological Status Monitor is being developed. Mini Mitter Company successfully developed a first version of the heat flux sensor/temperature sensor in conjunction with the University of Minnesota. A summary description of the first prototype follows. Temperature sensor - The temperature sensor we developed has been based on the VitalSense dermal patch thermometer design. We adapted the dermal patch circuit to utilize a remote thermistor whose thermal characteristics were identical to the thermistor used in our commercially-available dermal patch thermometer. The remote thermistor is a miniature cylindrical sensor having a lead length of about four inches. The thermistor is small enough to fit under a finger bandage or other securing device on the digit extremity. Because the thermistor has the same thermal characteristics as our dermal patch thermistor, there was no additional requirement for special calibration, etc. The remote thermometer could be used without any changes to the VitalSense system. Heat flux sensor - We used a conventional heat flux sensor having a voltage output proportional to the heat flux through an area approximately 0.25 inches square. We adapted a linear amplifier that had already been designed for our heart rate and activity sensors. The gain and offset of the amplifier was designed to match the characteristics of the sensor itself, and we balanced the output to be within the 3.0 volt window, at approximately 1.2 volts. Conversion of the amplified voltage was accomplished on board a circuit specifically designed to convert the voltage to a count value that could be recognized by the VitalSense monitor. In this way, we could record counts in the Monitor that are linearly related to the voltage output of the heat sensor. During this work, we noticed that outside air currents affect the stability of the measurement, so it is important to shield the sensor from any effects of ambient air currents. Calibration of Heat Flow Sensors The calibration of the temperature and heat flux sensors was carried out at the University of Minnesota Heat Transfer Laboratory, Department of Mechanical Engineering, and Department of Electrical Engineering. The aim was to calibrate two heat flow sensors (HFSs) against a known heat flux in a controlled environment. The HFSs calibration setup consists of a thermofoil heater sandwiched between two symmetric layers. Each layer consists of a HFS embedded within a piece of plexiglass, which is followed by a copper block. The heater is connected to a power supply and the thermal energy generated from passing a current through it produces a uniform heat flux on either surface. The entire setup is insulated using glass wool on the vertical sides and foam on the top and bottom surfaces. The HFSs are calibrated by measuring the output voltage corresponding to the heat flux (determined from the power input to the heater). Twenty thermocouples are embedded in the plexiglass pieces to estimate the temperature distribution to determine steady-state condition for voltage measurement from the HFS that also indicate the uniformity of heat transfer through the material. The calibration was performed in the flux range of 0-210 W/m2. The thermocouples (E-type) that were used in the temperature measurements were calibrated against a Standard Platinum Resistance Thermometer (SPRT). Five pieces of equal lengths were selected from the spool and the results showed a precision limit of 0.03oC based on the Students’ t-distribution within a 95% confidence limit. Runs in the HFS calibration range (0-210 W/m2) with black-black facing of the HFSs were also carried out. Calibration of Heat Flux Sensors to determine the value of Heat-flux (or correlation) versus Count value from the Vital Sense monitor. The heat passed through HF-1 was measured as emf across it, measured on a Keithley 2071 digital multimeter. The count value corresponding to the heat-flux passed through HF-2 was measured using the Mini Mitter Vital Sense monitor. Subsequently, the total heat flux, obtained by dividing the power by the cross-sectional area of the heater, is equal to the sum of the losses for each side. The heat-flux through HF-1 was determined from the output voltage/emf across HF-1, and as the calibration constant of HF-1 is known, its heat-flux can simply be calculated by C1 * V1, where C1 is the calibration constant of HF-1, and V1 is the corresponding output voltage measured across HF-1. Then, the heat-flux corresponding to the Vital Sense heat-flux is obtained by subtracting the heat-flux of HF-1 from the total heat-flux, which is calculated using Ohm’s law. In addition, since the Vital Sense monitor generates a count value in the range of 110 to 255 for heat-flux from the white-to-black-side, and 110 down to 0 for heat-flux from the black-to-white-side, the experiment was performed again, by placing the Vital Sense heat-flux sensor (HF-2) in both the positions. The voltage source is increased from 0 to 32 V such that a total heat flux from 0 to approximately 600W/m2 is generated from the heater. Output voltage of HF-1 and count-value of HF-2 is measured correspondingly. The experiment was repeated several times, to ensure that there was minimum deviation from optimum results. These calibration data were used by Mini Mitter to program the VitalSense monitor to display data from the finger calorimeter. Current work by Mini Mitter involves the miniaturization of the transmitter, which is worn on the wrist. Five laboratory experiments on 4 different subjects were conducted involving the application of local cooling on body areas remote (thigh) or close (forearm) to the finger calorimeter. The purpose was to assess whether these cold applications influence the accuracy of the information obtained by the finger calorimeter. The accuracy criterion was the quantity of heat insertion/extraction and relationship with core temperature and skin temperature topography. The findings demonstrated the high correspondence between the thermal status of the body core and the temperature/heat flux of the finger as registered by the finger calorimeter under local (thigh, forearm) cooling/warming.

	2. Assessment of the function of the Phase VI glove in cold conditions with and without insulation Four subjects donned in the SLCWG were tested wearing the Phase VI gloves with the entire micrometeoroid garment (TMG) on the left hand, and the glove without the TMG on the right hand. Inlet water temperature circulating through the SLCWG was lowered to 20ºC. Under the development of a general heat deficit/surplus in the body, the influence of differing insulation on each hand diminished. Therefore, the primary influence on local and total status is not distal thermal insulation, but the forming total deficit of heat in the body core organs. These findings have implications for the strategy of using physiologically designed gloves to increase blood circulation to the fingers, rather than increasing thermal insulation. 3. Evaluation of the operation of the finger calorimeter, Phase VI, and physiologically designed gloves while subjects grasp a highly cold rail Three subjects (2 males, 1 female) were studied over a total of 8 experimental sessions, wearing either the Phase VI gloves (Control) or physiologically designed gloves placed between the bladder and the TMG of the Phase VI glove (Experimental) while grasping a cold rail (-70oC). The protocol consisted of a baseline period followed by six 3 minute grasping intervals interspersed with 3 minute rest periods during which the subject placed his/her hands in their lap. The findings showed that in the Control condition, there was a large temperature fluctuation in the distal phalanx of the finger from 33oC (baseline) to 20oC (grasping interval), because of the low thermal insulation of the gloves. In the Experimental condition, with warm water (45 oC) circulating through the physiologically designed glove, the temperature/heat flux fluctuation was considerably less, 33 to 24oC. 4. Work focused on the physiologically designed warming gloves to identify the best type of wire to use as a heating element and the best glove fabric materials to provide optimal voltage to maintain glove temperature at 30oC during the imposition of cold to the body. 5. Three experiments conducted to further understand the efficacy of different garment tubing bypasses for emergency purposes – local cooling/warming of head/hand loop. 6. Successful completion of Ph.D. dissertation by project director Aitor Coca in June 2005 based on project data. Dissertation title: Control and Prediction of Body Thermal Status Through the Dynamic of Finger Heat Flux/Temperature.
Bibliography Type:	Description: (Last Updated: 02/07/2014)
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